

Longest Substring Without Repeating Characters

Given a string s , find the length of the longest substring without duplicate characters.

$s = \text{"abcabcbb"}$
o/p: 3

$s = \text{"bbbbbb"}$
o/p: 1

$s = \text{"pwwkew"}$
o/p: 3

Brute force

Try all possible substrings and check if each substring has duplicates $\rightarrow$ HashSet

TC: $O(N^2)$ SC: $O(N)$

Optimized maintain a (substring) window with unique char

"pwwkew"	↑↑↑	window	<u>pw</u> w	max = $\emptyset$
"pwwkew"	↑	window	<u>wkew</u>	1
"pwwkew"	↑	window	<u>kew</u>	2
				3

longestUniqueSubstring (s):

seen //set

start = 0, max = 0

//expand window using pointer

for end $\rightarrow$ 0 to s.length-1 :

//If duplicates found, shrink window

while s[end] is in seen:

remove s[start] from seen

start ++

add s[end] to seen

max = max(max, end - start + 1)

or seen.size()

return max

TC: $O(N)$

SC: $O(N)$